

RESEARCH ARTICLE

# Effects of cover soil stockpiling on plant community development following reclamation of oil sands sites in Alberta

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Stockpiling of cover soil can influence vegetation development following reclamation. Cover soil, comprising the upper 15–30 cm of the surface material on sites scheduled for mining, is commonly salvaged prior to mining and used directly or stockpiled for various lengths of time until it is needed. Salvaging and stockpiling causes physical, chemical, and biological changes in cover soils. In particular, stockpiling reduces the availability and vigor of vegetative propagules and seed, and can lead to increases in the abundance of some weedy species. This study uses data from monitoring plots to assess how stockpiling of cover soil impacts plant community development on reclaimed oil sands mine sites in northern Alberta. Development of plant communities differed distinctly between directly placed and stockpiled cover soil treatments even 18 years after reclamation. Direct placement of cover soil resulted in higher percent cover, species richness, and diversity. Nonmetric multidimensional scaling and multiresponse permutation procedure revealed compositional differentiation between the treatments. Indicator species analysis showed that direct placement treatment was dominated by perennial species while grasses and annual forb species dominated sites where stockpiled soil was used. Results indicate that stockpiling leads to slower vegetation recovery while direct placement of cover soil supports more rapid succession (from ruderal and annual communities to perennial communities). In addition, direct placement may be less costly than stockpiling. However, scheduling of salvage and placement remains a challenge.

**Key words:** disturbance, diversity and richness, oil sands reclamation, plant community, species co-occurrence

## Implications for Practice

- Direct placement of cover soil during reclamation can facilitate establishment of a plant community with many desirable native forest understory species.
- Although the trajectory of community development on the direct placement treatment follows the typical early successional progress of boreal forests, the community assembly process is unstructured and requires more time to form a stable plant community.
- It is very important to consider alternative options in stockpile management in order to maintain the viability of native seed and vegetative propagules.

## Introduction

Open pit mining of oil sands in the boreal forest in northeastern Alberta, Canada, is one of the world's largest and most intense anthropogenic disturbances. Oil sands operations have removed (as of 2016) nearly 953 km<sup>2</sup> of forests (complete removal of the forest canopy and understory vegetation) (Alberta Environment and Parks 2017). Long-term changes in soil chemistry, altered water and nutrient cycles, and significant topographical transformations to the original landscape are evident following these operations (Wickham et al. 2007; Saylor 2008). Government of Alberta policy requires that a disturbed mine site

should be reclaimed to equivalent land capability, as mandated by the Environmental Protection and Enhancement Act of Alberta (Alberta Government 2005; Oil Sands Revegetation Reclamation Committee 1998, Section 1(e) Alberta Regulation 115/93). The term equivalent land capability means “the ability of the land to support various land uses after conservation and reclamation is similar to the ability that existed prior to an activity being conducted on the land, but that the individual land uses will not necessarily be identical” (Alberta Environment 2010). Therefore, the common goal of restoration of local boreal ecosystems is the creation of diverse, self-sustaining, and resilient native communities (Alberta Government 2005). While seeding or planting nursery grown plants is possible, costs and limited supplies of seed are problematic (Macdonald et al. 2012, 2015a). Salvaged cover soil materials (e.g. peat soil

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from lowlands and forest floor soil from upland forests), which contains seeds and vegetative propagules, are recognized as significant resources for community establishment (MacKenzie & Naeth 2010; Macdonald et al. 2015b; Schott et al. 2016; Dhar et al. 2018a). Moreover, the seed reserve in cover soil is a source of inexpensive, diverse, and ecologically adapted vegetation (MacKenzie & Naeth 2007; Naeth et al. 2013) and, for some plants, it may be the only reserve of propagules from which a species can establish (Koch & Ward 1994; Ward et al. 1996). According to Ward et al. (1996) more than 70% of the species found on a reclaimed site originate from propagules contained in the salvaged cover soil. Results from various studies indicate that seed bank densities in the Canadian boreal forest cover soil range between 1,273 and 9,108 seeds/m<sup>2</sup> (Moore & Wein 1977; Hills & Morris 1992; MacKenzie & Naeth 2010). In addition, the use of salvaged cover soil can accelerate restoration of soil structure and function (organic matter content, nutrient concentrations, microorganisms, water-holding capacity) (Sydnor & Redente 2002). Consequently, salvaged cover soils are widely used in reclamation across the globe (Rokich et al. 2000; Zhang et al. 2013; Errington & Pinno 2016; Ulrich et al. 2016). Since the majority of the landscape around Alberta oil sands mining sites is covered by peat surface soils (i.e. “Muskeg”) this material is commonly used in mixture with mineral soil as cover soil and applied as “peat-mineral mix” (MacKenzie & Naeth 2010; Schott et al. 2016). The ratio of peat-mineral mix varies between 3:2 and 3:4 by volume (peat and mineral soil) and is placed in the reclamation sites at the final stage of operation (Alberta Environment and Water 2012; Dhar et al. 2018a).

During mining operations, the peat layer and underlying mineral substrates are commonly salvaged and used as cover soil directly or stockpiled and then placed at a later date (Abdul-Kareem & McRae 1984; Mackenzie & Naeth 2010). Cover soil has a tendency to change chemically, biologically and physically during stockpiling (Grant & Koch 2007; Mackenzie & Naeth 2010; MacKenzie et al. 2012). Some of these changes are due to the handling of the soil and others from the anaerobic conditions that may occur within the stockpiles (Abdul-Kareem & McRae 1984; Mackenzie & Naeth 2010; MacKenzie et al. 2012). Anaerobic conditions can occur due to increases in temperature, water content, carbon dioxide, methane, and ethylene; particularly in lower layers of deep stockpiles (Grant & Koch 2007; Mackenzie & Naeth 2010; MacKenzie et al. 2012). These changes can have detrimental effects on plant propagule viability and subsequent plant community development following placement on reclaimed sites (Mackenzie & Naeth 2010; MacKenzie et al. 2012). Some studies report that stockpiling rapidly reduces native seed viability (Dickie et al. 1988; Rokich et al. 2000; Rivera et al. 2012) with several studies showing decreased viability of propagules following stockpiling for as little as 10 months (Koch et al. 1996; Rokich et al. 2000; MacKenzie & Naeth 2009). While some studies have examined the impact of stockpiling on soil properties, propagule banks, and germination (Mackenzie & Naeth 2010; MacKenzie et al. 2012), the effect of stockpiling on plant community succession has received little attention.

The overall aim of this study was to enhance our understanding of how stockpiling of cover soils affects plant community development on reclaimed oil sands mine sites using long-term (18–24 years) monitoring data. Specifically, the goal of this study was to compare effects of direct placement of cover soil and placement of cover soil after stockpiling on plant community development in terms of species composition, richness, diversity, and species co-occurrence.

## Methods

### Site Description

This study utilized the Long Term Plot Network (LTPN) data collected from oil sands mining reclamation sites, located 30 km north of Fort McMurray, Alberta (56°59'N and 111°29'W) in the central mixed-wood natural subregion of the boreal forest region (Beckingham & Archibald 1996; Fung & Macyk 2000). The area is characterized by long cold winters and short warm summers with a mean temperature of 17.1°C in July and in –17.4°C in January, and mean annual precipitation of 419 mm, of which 134 mm falls as snow (Errington & Pinno 2016). The majority of soils have developed on glacial and glaciofluvial deposits. Gray Luvisolic soils (based on the Canadian System of Soil Classification (Soil Classification Working Group 1998)) are associated with till and lacustrine deposits, while Dystric Brunisols are associated with glaciofluvial outwash and eolian sands (Beckingham & Archibald 1996). Dominant tree species in natural forests of the region are trembling aspen (*Populus tremuloides*), balsam poplar (*Populus balsamifera*), paper birch (*Betula papyrifera*), white spruce (*Picea glauca*), black spruce (*Picea mariana*), jack pine (*Pinus banksiana*), and tamarack (*Larix laricina*), which grow in both pure and mixed stands (Downing & Pettapiece 2006).

### Sampling Design and Data Source

Eight study sites were selected from the LTPN established by the Cumulative Environmental Management Association (CEMA) (Table 1). CEMA was a multistakeholder group responsible for developing guidance documents to support reclamation in the area. Reclamation activities at these sites occurred between 1987 and 1993 or 18–24 years from data collection. The area of the reclaimed sites ranges from 2.4 to 7.6 ha (Table 1). All reclamation sites were reconstructed using overburden materials where peat-mineral mix was used as a cover soil in five different years (1987, 1988, 1990, 1992, and 1993). The peat soils were salvaged prior to mining and then directly placed on the reclaimed sites or stockpiled until sites were ready for cover soil placement. The depth of salvaged peat soils as well as the duration of stockpiling is unknown, but stockpiling is known to be more than 1 year; however, placement depth of the cover soil ranged from 16 to 33 cm (Table 1). Sites were grouped into two treatment categories: (1) direct placement of salvaged cover soil mix on overburden (five sites); and (2) placement of salvaged cover soil after stockpiling on overburden (three sites). Due to the lack of replication cover soil placement depth was

**Table 1.** Characteristics of different oil sand reclamation sites.

| Reclamation site | Reclamation year | Cover soil type  | Cover soil placement treatment | Area (ha) | Cover soil depth (cm) | Number of plots |
|------------------|------------------|------------------|--------------------------------|-----------|-----------------------|-----------------|
| Site-024         | 1987             | Peat-mineral mix | Direct                         | 2.9       | 21                    | 5               |
| Site-186         | 1990             | Peat-mineral mix | Direct                         | 3.6       | 20                    | 5               |
| Site-189         | 1993             | Peat-mineral mix | Direct                         | 6.1       | 33                    | 7               |
| Site-250         | 1991             | Peat-mineral mix | Direct                         | 4.5       | 20                    | 6               |
| Site-270         | 1992             | Peat-mineral mix | Direct                         | 5.9       | 29                    | 6               |
| Site-059         | 1988             | Peat-mineral mix | Stockpile                      | 4.9       | 20                    | 6               |
| Site-236         | 1990             | Peat-mineral mix | Stockpile                      | 2.4       | 16                    | 5               |
| Site-286         | 1992             | Peat-mineral mix | Stockpile                      | 7.6       | 29                    | 8               |

not included in the analysis. After cover soil placement, all eight sites were planted with white spruce (*P. glauca*), lodgepole pine (*Pinus contorta*), and trembling aspen (*P. tremuloides*). There were no records of seeding on the sites.

A grid pattern was used for the determination of plot locations within each reclamation site. The starting point for the grid was randomly chosen using ArcGIS and a uniform 100-m grid point interval was constructed in ArcGIS aligned to the cardinal directions (north, east, south, and west) and overlaid onto each reclamation site. Only the points located at least 20 m from the boundary of each site were used. Depending on the size of the reclamation site, between five and eight grid points were selected for establishment of plots. All vascular and nonvascular plants were identified to species level and cover was visually estimated in late summer 2011 for a 5 × 5 m (25 m<sup>2</sup>) established around the plot center. Nomenclature follows Moss (1983) for vascular plants and Anderson et al. (1990) and Brodo et al. (2001) for nonvascular plants (bryophytes and lichens). The total number and cover of nonnative species and rare species were also estimated. The total number of sample plots was adequate as species numbers reached a plateau for all treatment plots (Fig. S1, Supporting Information).

### Statistical Analysis

Vegetation cover percent and diversity indices (species richness, Shannon–Wiener’s index, and Shannon evenness) were used in the analysis for shrub, forb, graminoid, and bryophyte layers. Diversity indices were not analyzed for the tree layer because there were no plots with more than three tree species.

Linear mixed-effects models were used to examine effects of stockpiling on vegetation cover and diversity indices. The most appropriate model was selected based on the Akaike information criterion value where treatment (direct placement vs. stockpiling), reclamation year (1987, 1988, 1990, 1992, and 1993), and slope position of the sites were treated as fixed effects and plot was treated as a random factor. Analyses were conducted using the function “lme” from the package “nlme” in the R statistics system version 3.2.5 (R Core Team 2016). Tukey HSD post hoc comparisons ( $\alpha = 0.05$ ) were carried out using the function “glht” from the package “multcomp” 1.3-2 if significant differences were found. Normality and homogeneity of variance were tested by examining plots of residuals and the

normal q–q plots. Square root transformations were carried out when needed.

Nonmetric multidimensional scaling (NMDS) ordination was conducted to examine plant community composition and species assemblages as they relate to physical environmental attributes (McCune & Grace 2002). NMDS uses an iterative search to converge on a solution that minimizes “stress” in the configuration. Stress is a function of a linear monotone transformation of observed dissimilarities and ordination distances. The ordination was conducted using the package “vegan 2.4-2” (Oksanen et al. 2017) in the R statistics system. To support the descriptive ordination and to quantify multivariate differences in species composition between treatments, a nonparametric multivariate test “multiresponse permutation procedure (MRPP)” (McCune & Grace 2002) was applied. MRPP is useful with ecological community data that do not meet distributional assumptions required of parametric tests that address similar questions (McCune & Grace 2002). MRPP was conducted using “vegdist” within the package “vegan 2.4-2” (Oksanen et al. 2017).

Further, the affinity of species to particular treatment units was evaluated using indicator species analysis (Dufrêne & Legendre 1997; De Cáceres et al. 2012) on abundance data with the function “multipatt” of the package “indicspecies” version 1.7.5 for the R statistics system (De Cáceres 2013). Function ‘multipatt’ is the most commonly used function of “indicspecies”; it allows the determination of species lists associated with particular groups of sites (or combinations of those) (De Cáceres et al. 2012).

Species co-occurrence patterns were analyzed using the probabilistic model of species co-occurrence or probabilistic pairwise procedure (Veech 2013), which indicates the probability that two selected species co-occur at a frequency either less than or greater than would be expected to occur randomly. These probabilities can be obtained analytically under the condition where a species probability of occurrence at each site is equal to its observed frequency among all the sites. The probabilistic model employs combinatorics to determine the probability that the observed frequency of co-occurrence is significantly large and greater than expected (positive association), significantly small and less than expected (negative association), or not significantly different and approximately equal to expected (random association) (Veech 2013). The analysis was conducted

using the R package “co-occur 1.3” (Griffith et al. 2016); it produces, for all species pairs, exact probabilities of co-occurrence greater than or less than what is observed as well as the positive, negative, and random association of species.

## Results

The mixed model analysis indicated that total cover differed significantly between the treatments for trees ( $p < 0.001$ ), forbs ( $p < 0.05$ ), and graminoids ( $p < 0.001$ ) while reclamation year was significant for shrubs ( $p < 0.05$ ), graminoids ( $p < 0.05$ ), and bryophytes ( $p < 0.001$ ) (Table S1). The stockpiling treatment had lower percent cover for all layers except graminoid and bryophyte (Fig. 1). Graminoid and bryophyte cover in the stockpiling treatment was 49.5% and 15.3% compared to 12.2% and 12.0% in the direct placement treatment (Fig. 1). The direct placement treatment had more than double the cover (54.2%) of the stockpiling treatment (25.1%). Although shrub cover differed significantly among years there was no evident trend (Fig. S2). Post hoc pairwise comparisons revealed a significant difference between years 1987 and 1990 for shrub ( $p < 0.05$ ) and graminoid ( $p < 0.05$ ) cover. The bryophyte layer showed more prominent year effects with significant differences being evident for reclamation years 1987–1990 ( $p < 0.05$ ), 1987–1991 ( $p < 0.01$ ), 1988–1990 ( $p < 0.001$ ), and 1988–1991 ( $p < 0.05$ ).

In 2011, a total of 81 species were documented across treatments (Table S2), which included six tree species, 16 shrub species (14 in direct and nine in stockpiling), 27 forb species (23 in direct and 19 in stockpiling), 14 graminoids (14 in direct and 10 in stockpiling), 13 bryophytes (nine in direct and 11 in stockpiling), and five lichens. Direct placement had more species (58: native 49 and nine nonnative) than the stockpiling treatment (47: native 38 nonnative nine). The most commonly observed tree species were lodgepole pine and trembling aspen; prickly rose (*Rosa acicularis*) and Saskatoon (*Amelanchier alnifolia*) were the most common shrubs; white sweet clover (*Melilotus albus*) and wild strawberry (*Fragaria virginiana*) were the most common forbs; Rocky Mountain fescue (*Festuca saximontana*) and Western wheat grass (*Pascopyrum smithii*) were common graminoids; and purple horn-toothed moss (*Ceratodon purpureus*) and golden ragged moss (*Brachythecium salebrosum*) were common bryophytes. Eleven nonnative species were found across the sites with each treatment having nine species with *M. albus* (3.5% cover) being most common in the direct placement treatment and Alsike clover (1.3% cover) being most common in the stockpiling treatment. There was no significant difference in the number of nonnative species between treatments. Most of the species found in both treatments are considered to be early seral species (Table S2). For the small number of late seral species found, Western wood lily (*Lilium philadelphicum*) and common pink wintergreen (*Pyrola asarifolia*) were found only in the direct placement treatment and stair-step moss (*Hylocomium splendens*) was found in both cover soil treatments (Table S2).

Species richness and diversity (Shannon index) differed significantly between treatments and among reclamation years for

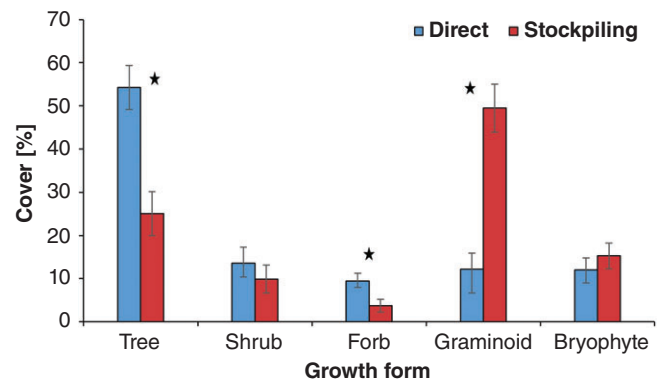


Figure 1. Bars illustrate mean ( $\pm$ SE) percent cover by topsoil placement treatments for different growth form (tree, shrub, forb, graminoid, and bryophyte) at 5  $\times$  5-m plots in Athabasca oil sands reclamation sites (asterisk indicates significant difference).

shrub (richness:  $p < 0.01$  and  $p < 0.01$ ; Shannon:  $p < 0.05$  and  $p < 0.05$ ) and forb layers (richness:  $p < 0.05$ , Shannon:  $p < 0.05$  and  $p < 0.01$ ) (Table S1). Shannon evenness differed significantly between treatments for the shrub layer ( $p = 0.024$ ) only and reclamation year was significant for the shrub ( $p < 0.05$ ), forb ( $p < 0.05$ ), and graminoid ( $p < 0.001$ ) layers (Table S1). Direct placement had greater richness and diversity (both Shannon index and Shannon evenness) across all layers except graminoid where the stockpiling treatment had slightly higher species richness (Fig. 2). For all layers combined, species richness and Shannon index were significantly larger for direct placement (Fig. 2). Richness, Shannon index, and Shannon evenness of shrubs, forbs, and graminoids differed significantly between years. Shrub richness differed between the 1987 and 1990 ( $p < 0.05$ ) and between 1987 and 1991 ( $p < 0.01$ ); Shannon index between 1987 and 1991 ( $p < 0.05$ ) and between 1991 and 1993 ( $p < 0.05$ ); and Shannon evenness between 1988 and 1991 ( $p = 0.008$ ). Nonnative species richness and diversity did not differ between treatments or years.

The NMDS ordination showed some level of separation between the soil placement treatments (Fig. 3) and yielded a two-dimensional solution, which explained 64.8% of the total variation. This implies that cover soil placement treatments influenced species composition, which was supported by MRPP analysis. The MRPP analysis indicated a significant difference in plant community composition ( $p < 0.05$ ,  $T = -2.219$ ,  $A = 0.291$ ) between treatments but indicated that differences between reclamation years were not significant. Indicator species analysis shows that more species were associated with direct placement (five species: Balsam poplar [*Populus balsamifera*], Lindley's aster [*Symphyotrichum ciliolatum*], common fireweed [*Chamaenerion angustifolium*], *M. albus*, and common dandelion [*Taraxacum officinale*]) than the stockpiling treatment (three species: Rocky Mountain fescue, northern brome [*Bromus inermis*] and Alsike clover [*Trifolium hybridum*]). Two nonnative indicator species were found in each treatment (for direct treatment: *M. albus* and *T. officinale* and for stockpile: *B. inermis* and *T. hybridum*) (Table S3). Structurally, one tree and four forb indicator species were found in



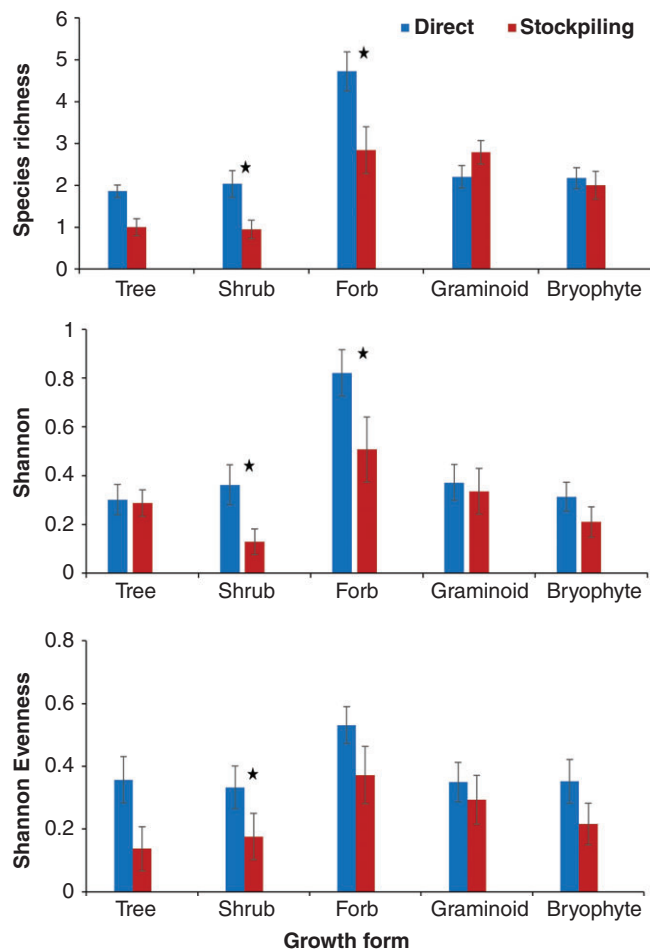


Figure 2. Bars illustrate mean ( $\pm$ SE) species richness and diversity by topsoil placement treatments for different growth form (tree, shrub, forb, graminoid, and bryophyte) at  $5 \times 5$ -m plots in Athabasca oil sands reclamation sites (asterisk indicates significant difference).

the direct placement treatment, while two graminoid and one forb species were indicative of the stockpile treatment. In addition, 25 species had the highest indicator value components and were found in both treatments (Table S2). When we conducted the indicator species analysis by reclamation years four perennial species (*Agropyron* sp. [nonnative graminoid], *Aster ciliolatus* [forb], *Equisetum arvense* [forb], and *Potentilla fruticosa* [shrub]) were associated with reclamation year 1987 and one species each for reclamation years 1988 (nonnative forb: *T. hybridum*) and 1993 (*P. balsamifera* [tree]) (Table S3).

Analysis of species co-occurrence considered 3,240 pairwise combinations of species where 2,129 or 65.7% of species pairs were removed from the analysis because the expected co-occurrence was  $<1$  (share less than one site). Results suggest that most of the classifiable species pairs had “truly random” associations (896 pairs). Only 1.4% of the potential pairwise associations were significant which is much lower than expected by chance (5%). The total number of positive associations was greater than the number of negative associations (10 positive compared to six negative; Fig. 4). All six

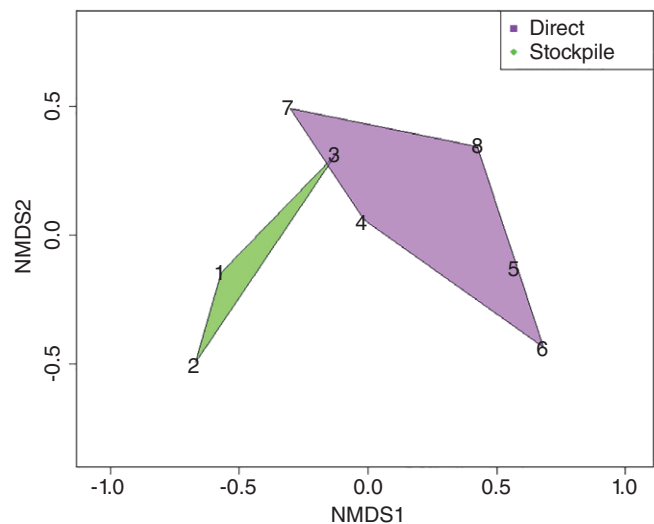


Figure 3. Nonmetric multidimensional scaling (NMDS) ordination of plant community by topsoil placement treatments and reclamation years in Athabasca oil sands reclamation sites (numbers indicate the reclamation years chronologically, for stockpiling: 1 = 1988, 2 = 1990, 3 = 1992; and for direct: 4 = 1987, 5 = 1990, 6 = 1992, 7 = 1992, 8 = 1993).

negative associations occurred with two tree species (white spruce and balsam poplar), while yellow sweet-clover (*Melilotus officinalis*) was the only forb species which showed both negative and positive association with balsam poplar and hybrid poplar (*Populus × populus*), respectively. Analysis of species co-occurrence by treatments and reclamation years showed no positive or negative pairwise species associations, indicating that cover soil placement treatment and reclamation years did not influence species associations measured 18–24 years after reclamation in these sites.

## Discussion

This study represents one of the first attempts to examine how stockpiling affects vegetation dynamics of reclaimed oil sand sites in Alberta. During the 18–24 years following reclamation, development of plant communities differed distinctly between the direct placement and stockpiling treatments. Other than planting of white spruce, lodgepole pine, and trembling aspen into the reclaimed uplands, plants establishing on the site were recruited into the area via dispersal of propagules from other habitats or were transferred to the site within the applied cover soil. Although there is no information on the causes of reduced seed viability during stockpiling, some studies suggest that in situ germination, predation, physical and mechanical damage, seed decomposition, and loss of microbial associates are the major reasons for loss of viable propagules in stockpiles (Dickie et al. 1988; Rokich et al. 2000; MacKenzie et al. 2012; Rivera et al. 2012; Naeth et al. 2013). According to Rokich et al. (2000) declines in seed viability are due to high temperature and water content. Dickie et al. (1988) suggest that anaerobic conditions deep in stockpiles could reduce seed viability, while MacKenzie

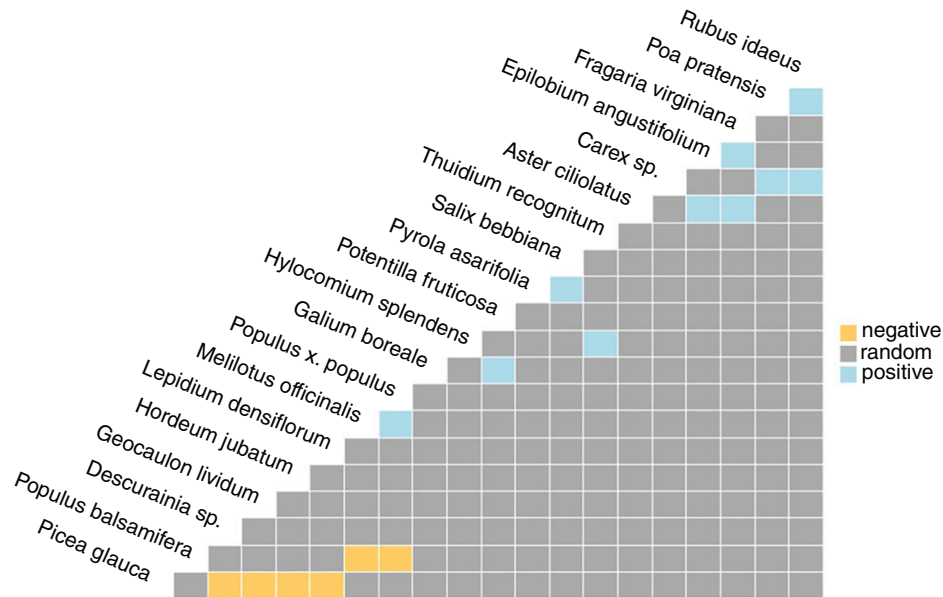


Figure 4. The potential significant pairwise associations based on probabilistic model of species co-occurrence analysis.

et al. (2012) state that propagule viability declined in both aerobic and anaerobic conditions. In the same study, MacKenzie et al. (2012) report that stockpiling of organic soil horizons significantly reduced seed viability for 24 of 27 boreal species in small and large stockpiles at depths below 1 m. Other studies also report reductions in native plant propagule viability during stockpiling (Rokich et al. 2000; Rivera et al. 2012; Naeth et al. 2013).

Duration of stockpiling can also influence vegetation development following placement of material on a reclamation site (Koch et al. 1996; Rokich et al. 2000; MacKenzie & Naeth 2009). MacKenzie et al. (2012) report that emergence of bearberry (*Arctostaphylos uva-ursi*), wild lily-of-the-valley (*Maianthemum canadense*), and common blueberry (*Vaccinium myrtilloides*) declined by more than 80% after 8 months and that no roots survived after 16 months. In another study, Naeth et al. (2013) found that loss of propagule viability occurred more slowly in winter-constructed stockpiles but after 12 months results were the same regardless of the season of stockpile construction in oil sands reclamation. Studies of other reclamation practices (such as bauxite mines or sand extraction restoration sites) also mentioned that 10 months stockpiling of salvaged cover soil could significantly decrease the viability of propagules (Iverson & Wali 1982; Koch et al. 1996; Rokich et al. 2000).

In contrast to stockpiling, this study indicates that direct placement of cover soil produces higher vegetation cover, diversity, and species richness for all vegetation layers except graminoid. Although the propagule bank was not examined in this study our results are consistent with previous work by Naeth et al. (2013), which indicated that direct application of cover soil resulted in greater native species abundance and diversity than the application of stockpiled cover soil. Stockpiling reclaimed sites had less than half of tree cover and almost four times

higher graminoid cover than those where cover soils had been placed directly. This implies that stockpiling of cover soil may lead to increases the abundance of grass species, which compete with trees and often leads to reduced tree cover. Most of these grass species are native species except two nonnative species (*Agropyron* sp. and *Bromus inermis*), which disperse from the surroundings. The nonnative species *B. inermis* may disperse into sites from adjacent reclamation areas where it is used for erosion control (Purdy et al. 2005). Some other studies also reported that stockpiling topsoil, even for short periods, substantially decreases seedling recruitment and species richness in bauxite mine restoration sites in Australia (Tacey & Glossop 1980; Ward et al. 1996). In another study, Rokich et al. (2000) reported a significant decline in species richness as the time of topsoil stockpiling increased in an Australian sand restoration site. From the above discussion, it can be concluded that the higher species richness and diversity in the direct placement treatment likely results from higher quality and survival of seed and other propagules.

Our indicator species analysis also showed that the perennial community dominated the direct placement treatment, while grasses and annual forb species dominated the stockpiling treatment. The functional groups to which the indicator species belong is an important aspect of characterizing plant communities. Trees and shrubs shade out ruderal species (Messier et al. 1998; Lieffers et al. 1999), which in turn can reduce competition and promote the development of forest understory species. Thus, the trajectory of community development following the direct placement treatment appears to be consistent with the typical successional progress of boreal forests from ruderal and annual communities to perennial communities (Landhäusser et al. 1996; Messier et al. 1998; Lieffers et al. 1999; Hart & Chen 2006; Dhar et al. 2018a). Conversely, grasses limit the establishment and development of tree and understory species

due to high competition (Landhäusser et al. 1996; Hart & Chen 2006) and thus stockpiling leads to slower successional progress and delayed ecosystem recovery. It is also important to recognize that plant communities in the stockpiled treatment will likely change over time when tree and shrub seedlings exert greater influence and propagules that disperse from surrounding areas start to establish (DeGrandpré et al. 2003; Hart & Chen 2006).

In this study, both direct placement and stockpiling treatments had less than 5% nonnative species cover even 19–24 years after reclamation, which is lower than reported by other oil sand reclamation studies in Alberta (Pinno & Hawkes 2015; Dhar et al. 2018b). Based on a study by Dhar et al. (2018b), nonnative species cover declined from 32 to 10% in 16–24 years after reclamation. Ongoing research with continuous monitoring is needed to focus on the potential long-term impacts of nonnative species on the ecosystem recovery of reclaimed sites.

Very limited positive and negative co-occurrence patterns in our study indicate largely random plant community assembly with reclaimed ecosystems being unstructured—24 years after reclamation. A similar plant community assembly pattern was also reported on open-cast lignite mine restoration in northeastern Germany (Zaplata et al. 2013; Ulrich et al. 2016), on a glacier outwash plain sites restoration in Iceland (Marteinsdóttir et al. 2018), and oil sand reclamation sites in Alberta (Dhar et al. 2017, 2018b). Based on the studies by Dhar et al. (2017, 2018b) plant community assembly in oil sands reclamation sites showed a random pattern of community assembly when peat–mineral mix was used as a cover soil. Similarly, Zaplata et al. (2013) reported that plant community succession was dominated by random patterns of species co-occurrence in the initial year of lignite mine restoration. According to Ulrich et al. (2016) initial species segregation showed a trend toward random co-occurrences in null model analysis. This suggests that during earlier stages of plant community assembly there are less competitive and inhibitory effects of one species on another (Tilman & Pacala 1993; Tilman 1996; Hubbell 2001; Fargione et al. 2004) and, as a result, existing species allow establishment of new species that have similar resource requirements (Gleason 1926). Mining and reclamation processes severely alter soil properties and biological legacies (i.e. surviving trees, snags and logs, patches of intact vegetation, and seed banks in tree crowns or in the soil), which negatively affect the memory of the ecosystems genetic, compositional, and structural conditions prior to disturbance (Drever et al. 2006; Dhar et al. 2016; Dhar et al. 2018a). Moreover, slow rates of establishment, growth, and development of species following reclamation in this cool boreal climate can result in availability of unused resources and space, enabling invasion of new species in the reclaimed ecosystem. While this may reflect incomplete utilization of resources in developing ecosystems, further investigation is recommended and should include monitoring of plant community assembly processes and resource availability over extended periods for different cover soil types (e.g. forest floor–mineral soil, peat–mineral soil, and so on) in comparison with naturally disturbed (e.g. fire) sites.

The expression of the seed bank and vegetative propagules from salvaged cover soil is influenced by salvage depth, stockpiling duration, and depth of cover soil placement (Rokich et al. 2000; MacKenzie et al. 2012; Macdonald et al. 2015b). Size, depth, and duration of stockpile also influence the survival of propagules stored in the soil seed banks (Mackenzie & Naeth 2010; MacKenzie et al. 2012). Several parameters that could be used to characterize of substrate and environmental conditions such as water holding capacity, soil water content, soil temperature, and chemical properties (available nutrients and soluble ions), and duration of the stockpiling and microbial communities were not available for the sites examined in this study. This limits our ability to examine factors which may be contributing to observed outcomes. Future studies need to consider substrate and environmental factors when examining the effects of stockpiling and direct placement of cover soil in reclaimed boreal forests.

While this study shows that stockpiling has detrimental impacts on plant community development, it is not always possible to directly place salvaged cover soil. Therefore, it is very important to consider alternative options in stockpile management in order to maintain the viability of native seed and vegetative propagules. This could include keeping the storage time of stockpiling as short as possible, and disturbance of cover soil (during salvaging and storing in stockpiles and removal and application to reclaimed sites) to a minimum level. In addition, weed control is important at all stages of the process; this should include control in adjacent areas as well as within stockpiles.

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## Supporting Information

The following information may be found in the online version of this article:

**Table S1.** Results of mixed-effects models for cover and diversity of different vegetation growth form in oil sand reclamation sites.

**Table S2.** Species found in the oil sand reclamation areas with indicator species analysis in two cover soil placement treatments.

**Table S3.** Significant ( $p < 0.05$ ) indicator species for the cover soil replacement treatments in oil sands reclamation sites.

**Figure S1.** Species accumulation curves in oil sands reclamation sites showing number of species (mean  $\pm$  SD) versus number of plots ( $5 \times 5$  m).

**Figure S2.** Shrub cover by reclamation years and treatments.

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